Product Requirements Document (PRD)

Product Name

VectorShift Pipeline Builder Enhancement

Product Overview

A visual workflow builder where users can create pipelines by connecting nodes together. Users can add different node types, configure node settings, connect nodes using edges, and submit the workflow for backend validation.

The application should provide a clean, intuitive node-based interface similar to modern AI workflow builders.

Problem Statement

Current implementation has:

- Repetitive node code
- Minimal styling
- Limited Text Node functionality
- No complete backend integration
- Poor scalability when adding new node types

The goal is to improve maintainability, usability, and functionality.

Target Users

Primary Users

- AI Engineers
- Workflow Designers
- Developers

Secondary Users

- Product Managers
- Automation Builders

User Roles

Workflow Creator

Can:

- Add nodes
- Configure nodes
- Connect nodes
- Submit workflows

Core Features

Node Management

- Create nodes
- Delete nodes
- Connect nodes
- Configure nodes

Dynamic Text Node

- Auto-resizing text area
- Dynamic variable detection
- Dynamic handle generation

Pipeline Analysis

- Count nodes
- Count edges
- DAG validation

Modern UI

- Consistent node design
- Responsive layout
- Better visual hierarchy

User Stories

US-01

As a user, I want to create a workflow by dragging and connecting nodes.

US-02

As a user, I want node types to have a consistent design.

US-03

As a user, I want text nodes to grow automatically when content increases.

US-04

As a user, I want variables like {{input}} to automatically create connection handles.

US-05

As a user, I want to submit my workflow and receive validation results.

US-06

As a user, I want to know whether my workflow is a DAG.

Success Metrics

- New node creation requires minimal code.
- Dynamic handles generate correctly.
- Pipeline submission succeeds.
- DAG validation works accurately.
- UI is visually appealing and consistent.

MVP Scope

Included:

- Base Node abstraction
- Existing nodes migrated
- Five additional nodes
- Dynamic Text Node
- Backend integration
- DAG validation
- Styled UI

Excluded:

- Authentication
- Persistence

- Collaboration
- Real execution engine
- Versioning